CrackDistill: Distilling SAM 2 into YOLOv11n-seg for Crack Segmentation

Progress report (June–October 2026), findings, and next architectures

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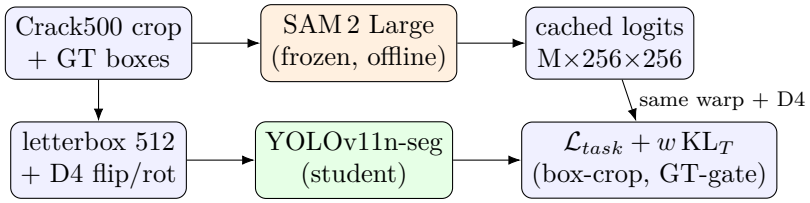
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Abstract

We distil the Segment Anything Model 2 (SAM 2.1 Hiera-Large, 224M parameters) into a 2.84M-parameter YOLOv11n-seg student for pavement-crack instance segmentation on Crack500. Over five months (79 commits) the project went from early KD "gains" to a corrected, controlled protocol. After fixing several silent bugs (teacher misalignment of about 140 px, wrong instance pairing for 58% of pairs, a baseline that still received KD loss), we find that **the KD gain seen without augmentation (+0.053 mask mAP50) disappears once the baseline is properly augmented**. Flips and rotations alone add +0.104, and standard YOLO augmentation is best overall (val mask mAP50 0.540; test pixel F1 0.731 on crops, 0.707 on full photos). A data-efficiency study (50% of the data) also shows no KD benefit. We review 2023–2026 literature and propose next steps: a better teacher (crack-fine-tuned SAM), SAM 1+SAM 2 logit fusion with standardisation, and KD schedules.

1 Goal and setup

- **Task:** crack instance segmentation; a real-time student for edge devices.
- **Teacher:** SAM 2.1 Hiera-L, prompted with each ground-truth crack's box. Pre-sigmoid mask logits are cached offline as `M×256×256` per image (2,196 files).
- **Student:** YOLOv11n-seg (2.84M parameters; 10.8 ms per 512 tile on a T4). It beat YOLOv8n-seg in early tests (0.540 vs 0.532).
- **Data:** Crack500, official split (1,896 / 348 / 1,124 crops; 200 full test photos).
- **KD loss:** the task loss plus a temperature-scaled Bernoulli KL between student mask logits and the teacher logits, warped into the student's letterboxed frame.



2 Evolution of the project

P0 (Jun–mid Jul). SAM 2 logits cached (15 Jun). First YOLOv8 KD notebooks; FP16 NaN in the KD loss fixed by clamping.

P1 (20 Jul–4 Aug). Repository created. Optuna tuned the loss weights (T=3.78, w_{mask}=0.96). Box and box+centroid prompts gave identical results. YOLOv11n chosen as the student.

P2 (5–14 Aug). Ablations: mask-KL was the useful term and feature MSE hurt. We found that some KD notebooks had *never loaded teacher logits*; every run now prints a logit-file count.

P3 (15 Aug–11 Sep). Many variants: dilated, affinity, multi-scale, channel-wise feature KD (CWD), Tversky, two-stage, out-of-distribution tiling. All scored 0.53–0.55 mask mAP50, and a 21-page draft was written. These numbers were later invalidated.

P4 (16–29 Sep): scientific audit. Bugs found and fixed:

- the training-time letterbox pad was read as 0.8 px instead of 112 px, so the teacher sat about 140 px off the crack;
- `overlap_mask` reordered instances, so only 42% of teacher–student pairs were correct (99.9% after the fix);
- the "clean" baseline still received the KD loss;
- the baseline had augmentation and the KD arms did not;
- the GT-soft control actually used SAM logits;
- EXIF rotation was handled wrongly (130 of 200 test photos);
- the seed was not passed to Ultralytics, feature hooks sat on the wrong layers, and the Kaggle dataset linker was ambiguous.

The measured SAM-vs-GT median IoU is 0.605.

P5 (early Oct): v1 protocol, no spatial augmentation. Baseline 0.410, Mask-KD 0.463, GT-soft 0.414. Every arm overfit (best epoch 9–17).

P6 (14 Oct): v2 design. Augmentation-consistent KD: the cached teacher map receives the same D4 flip/rotation as the image. Our measurements showed the crack fills only 0.9% of the KD canvas and SAM recalls only 56% of cracks 4 px wide or thinner, which motivated a new box-cropped, GT-gated KL (A5). The feature, boundary and affinity losses were dropped.

P7 (24 Oct): Stage A and **P8 (28 Oct): data-efficiency**, below.

3 Results

Table 1: Stage A, seed 0, Kaggle T4, 150 epochs. Val = 348 crops; test = scale-matched evaluator (thresholds tuned on val).

Arm	val mask mAP50	test crops F1	200 full photos F1	T4 time
v1 baseline (no spatial aug.)	0.410	–	–	1.2 h
v1 Mask-KD (no spatial aug.)	0.463	–	–	2.0 h
A1 baseline + D4	0.514	0.713	0.684	1.4 h
A2 Mask-KD + D4	0.503	–	–	2.0 h
A3 GT-soft control + D4	0.481	–	–	2.4 h
A5 gated Mask-KD + D4	0.511	0.704	0.661	2.6 h
A4 baseline + YOLO default aug.	**0.540**	**0.731**	**0.707**	1.4 h

Table 2: Data efficiency (val mask mAP50). Nested whole-photo subsets, identical for both arms.

Training data	Baseline (D4)	Gated KD (D4)	Δ
25% (478 crops)	0.454	pending (rerun)	–
50% (931 crops)	0.502	0.482	−0.020
100% (1,896 crops)	0.514	0.511	−0.003

Findings.

1. **The v1 KD gain was a substitute for augmentation.** D4 alone added +0.104, twice the KD gain, and KD added nothing on top.
2. **SAM is a noisy teacher on thin cracks.** It misses 44% of thin-crack pixels, and 21% of its crack pixels are ground-truth background.
3. **Full photos must be tested at the training scale.** With an older checkpoint, ×0.8 tiling scored F1 0.71 against 0.60 when the whole photo was squeezed into 512 px.
4. **Deliverable model:** A4, with 2.84M parameters and about 11 ms per tile.

Limitation: one seed so far. The claim is "no gain", not "harm".

4 Diagnosis: the zero-shot teacher problem

Knowledge distillation only helps when the teacher knows something the student cannot learn from the labels. Our measurements show that zero-shot SAM 2 does not:

- **The teacher is worse than the labels.** SAM 2 against the ground truth: median IoU 0.605. Recall is 0.56 on cracks 4 px wide or thinner, and 21% of its crack pixels are ground-truth background.
- **The student alone already reaches pixel F1 0.71–0.73.** A teacher that is not clearly better only adds noise or label smoothing. The GT-soft control (A3) behaves the same way.
- **The KD gain was really augmentation.** v1 KD gave +0.053 without spatial augmentation. Flips and rotations alone gave +0.104, and KD then added nothing.
- **The cached teacher cannot follow mosaic or scale augmentation**, which is exactly where the best model (A4) gains.
- **Fusing two zero-shot teachers (SAM 1+SAM 2) is unlikely to help by itself.** Both are prompted with the same boxes and both miss thin cracks, so their errors are correlated.

Implication: the bottleneck is the teacher's quality, not the KD loss. The teacher must first be made better than the student, for example a crack-fine-tuned SAM with val pixel F1 above 0.80. Alternatively, it should be used where no labels exist (pseudo-labelling unlabelled images).

5 Paper positioning

The work done so far supports a paper whichever way the next experiments go:

- **Negative result (already supported).** "Zero-shot foundation-model KD gains on a tiny crack segmenter are an artefact of weak baselines." Evidence: the audit and its bug list (§2), the GT-soft control, matched augmentation, the data-efficiency curve, and scale-matched full-photo evaluation. Many KD papers compare against unaugmented baselines, as our own v1 did, so this is a useful and honest contribution. What it still needs: 3 seeds and the 25% point.
- **Positive result (target).** "A crack-tuned SAM teacher with semi-supervised distillation improves a 2.8M-parameter segmenter beyond strong augmentation." Requirement, fixed in advance: beat A4 by at least 0.02 on val and test, over 3 seeds.
- **Either way:** also report out-of-distribution results (DeepCrack, CFD), where a foundation-model teacher may help generalisation even without an in-domain gain.

6 Related work (2023–2026) and what it implies

- **SAM is a noisy teacher on thin structures.** Boxes2Pixels [1] uses box-prompted SAM masks for defects and adds online self-correction for SAM's false negatives (+9.7 binary IoU).

SAMRefiner [2] cleans SAM pseudo-masks (+10%). CoGenSAM [3] and SERD [4] report that raw SAM masks drop or over-segment cracks, and that SAM's internal response keeps more crack pixels than its final masks (recall 82.7% vs 74.7%).

- **A better teacher.** CrackSAM [5] (adapter/LoRA), Segment Any Crack [6] (fine-tuning only the normalisation layers), and SAM2-Adapter [7] and SECrackSeg [8] all fine-tune SAM for cracks. A crack-tuned teacher is the most likely route to a real gain.
- **Multi-teacher fusion.** Logit standardisation [9] z-scores each teacher's logits, which is needed before averaging SAM 1 and SAM 2. Adaptive per-pixel teacher weights for segmentation [10]. AM-RADIO [11] uses a separate head per teacher instead of averaging. TOM [12] distils several teachers, including SAM, into one student. DIST [13] handles a large teacher–student capacity gap.
- **KD and augmentation.** The teacher must be consistent with the augmented view [14]. KD is partly label smoothing [15, 16], and stronger augmentation combined with plain KD is competitive [17].
- **Semi-supervised use of SAM.** SemiSAM [18], S4M [19] and SAM pseudo-labels inside UniMatch [20] turn SAM into extra supervision on unlabelled images. This is where a teacher can help when labels are scarce.
- **Lightweight crack models.** SCSegamba [21] (CVPR 2025) is a strong lightweight baseline to compare with.

7 Proposed next architectures

7.1 SAM 1 + SAM 2 teacher fusion

Cache SAM 1 (ViT-H) box-prompt logits $z^{(1)}$ next to the existing SAM 2 logits $z^{(2)}$ (about 1–2 T4-hours). Fuse per pixel after standardisation:

$$\tilde{z}^{(k)} = \frac{z^{(k)} - \mu_k}{\sigma_k}, \qquad z^\star = \sum_k \alpha_k(x)\, \tilde{z}^{(k)}, \quad \alpha_k(x) \propto \exp\big(-H(\sigma(\tilde{z}^{(k)}(x)))\big),$$

where H is the binary entropy, so the more confident teacher dominates at each pixel. Three variants: **(F1)** a plain mean; **(F2)** entropy weighting; **(F3)** agreement gating: KD only where both teachers agree with each other and with the ground truth, and the ground truth elsewhere. Plus an AM-RADIO-style option, **(F4)**: two lightweight mask-coefficient heads, one per teacher, removed at inference.

7.2 When to mix the teacher into YOLO

S1 Offline targets (current). KD at every step with a constant weight.

S2 Annealed KD. $w(e) = w_0 \cos^2(\pi e/2E)$. In our logs KD slowed the late-epoch decline, so it should be strong early and fade late.

S3 KD for the first N epochs, then the task loss only. A cheap test of the "early regulariser" effect.

S4 Pseudo-label refinement. A fused SAM mask, refined and gated by the ground truth, replaces the hard labels once before training.

S5 Online teacher. The teacher is run on the augmented view itself (SAM at train time, $\sim$3$\times$ slower). This lets KD use mosaic and scale augmentation, which is where A4's advantage comes from.

7.3 Priority plan (about 15 T4-hours)

1. Rerun E2 (25%, KD) to complete the curve; 2 more seeds of A1/A5 to get error bars.
2. Teacher upgrade: fine-tune the normalisation layers of SAM 2 on Crack500 train, re-cache the logits, and repeat A5 (most promising per the literature).
3. SAM 1+SAM 2 fusion F2/F3 with schedule S2, under A4-style augmentation for the baseline as well.
4. Evaluate on unseen datasets (CFD, DeepCrack) with pixel F1 and clDice.

Success criterion, fixed in advance: beat A4 by at least 0.02 val mask mAP50, with the same sign in test pixel F1.

References

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